

# Intersections of longest cycles and paths in $k$ -connected graphs and vertex-transitive graphs

Ziyuan Zhao

University of Science and Technology of China

[zyzhao2024@mail.ustc.edu.cn](mailto:zyzhao2024@mail.ustc.edu.cn)

*joint work with Jie Ma.*

December 4, 2025

# Overview

This talk involved the length/intersection properties of **longest cycles** in:

- $k$ -connected graphs;
- vertex-transitive graphs.

# A Conjecture on longest cycle intersection

## Conjecture (Smith)

For any  $k \geq 2$ , every two longest cycles in a  $k$ -connected graph must intersect in at least  $k$  vertices.

This has been verified for  $k \leq 8$  and  $k \geq (n + 16)/7$ .

## Theorem (Yepremyan et al., 2024.)

*Every two longest cycles in a  $k$ -connected graph intersect in at least  $\Omega(k^{\frac{5}{8}})$  vertices.*

# Main theorem 1

## Main theorem 1 (Ma, Z. 2025+)

For each two longest cycles in a 2-connected graph  $G$  intersecting in  $m$  vertices, there is a vertex-cut of size  $O(m^{\frac{3}{2}})$  separating them.

## Corollary (Ma, Z. 2025+)

Every two longest cycles in a  $k$ -connected graph intersect in at least  $\Omega(k^{\frac{2}{3}})$  vertices.

The main lemma has other applications, for such a separator is a "transversal".

# Transversals

## Definition

For a graph  $G$ , a **transversal** of  $G$  is a vertex set that intersects every longest cycle in  $G$ .

## Fact

In any 2-connected graph  $G$ :

- Every two longest cycles in  $G$  must intersect.
- The separator of every two longest cycles in  $G$  is a transversal.

# A problem on transversal

## Open question (Gallai)

Whether every connected graph  $G$  has a transversal of size  $10^{10}$ ?

## Theorem (Norin, Steiner, Thomassé, and Wollan, 2025)

*Every connected  $n$ -vertex graph  $G$  has a transversal of size  $O(\sqrt{n})$ .*

## Main theorem 1 (Ma, Z. 2025+)

For each two longest cycles in a 2-connected graph  $G$  intersecting in  $m$  vertices, there is a vertex-cut of size  $O(m^{\frac{3}{2}})$  separating them.

## Corollary

Let  $m$  be the minimum size of intersection between two longest cycles in  $G$ , then  $G$  has a transversal on  $O(m^{\frac{3}{2}})$  vertices.

# A Conjecture on cycle lengths in vertex-transitive graphs

## Definition

A graph  $G = (V, E)$  is *vertex-transitive* if its automorphism group  $\text{Aut}(G)$  acts *transitively* on the vertex set  $V$ . That is, for every pair of vertices  $u, v \in V$ , there exists a graph automorphism  $\varphi \in \text{Aut}(G)$  such that  $\varphi(u) = v$ .

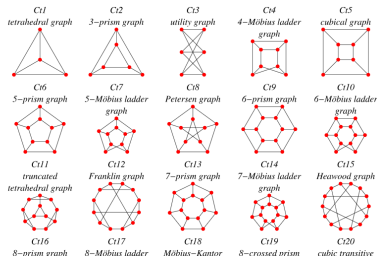


Figure: Cubic vertex-transitive graphs

# A Conjecture on cycle lengths in vertex-transitive graphs

## Conjecture (Lovász/Thomassen)

Every connected vertex-transitive graph on  $n \geq 3$  vertices contains a hamiltonian path/cycle, with only finitely many known exceptions.

## Theorem (Norin, Steiner, Thomassé, and Wollan, 2025.)

*Every connected vertex-transitive graph on  $n \geq 3$  vertices contains a cycle of length  $\Omega(n^{\frac{9}{14}}) \approx \Omega(n^{0.642})$ .*

# Some corollaries

## Fact

For a  $d$ -regular vertex-transitive graph  $G$  on  $n$  vertices:

- $G$  is  $2d/3$ -connected;
- (Devos) If every two longest cycles in  $G$  intersect in  $\Omega(m)$  vertices, then the longest cycle in  $G$  has length  $\Omega(\sqrt{mn})$ .

## Corollary (Ma, Z. 2025+)

Every connected  $d$ -regular vertex-transitive graph on  $n \geq 3$  vertices contains a cycle of length  $\Omega(d^{\frac{1}{3}} n^{\frac{1}{2}})$ .

This result is valid (relative to the bound  $n^{\frac{9}{14}}$  and  $d$ ) when  $\Omega(n^{\frac{3}{7}}) \leq d \leq O(n^{\frac{3}{4}})$ .

# Proof ideas

## Main theorem 1

Let  $X$  and  $Y$  be two longest cycles in a 2-connected graph  $G$  intersecting in  $m$  vertices. Then there is a vertex-cut of size  $O(m^{\frac{3}{2}})$  separating them.

Decompose  $X - Y$  (removing all vertices of  $Y$  from  $X$ ) and  $Y - X$  into  $m$  subpaths  $X_1, \dots, X_m$  and  $Y_1, \dots, Y_m$  respectively.

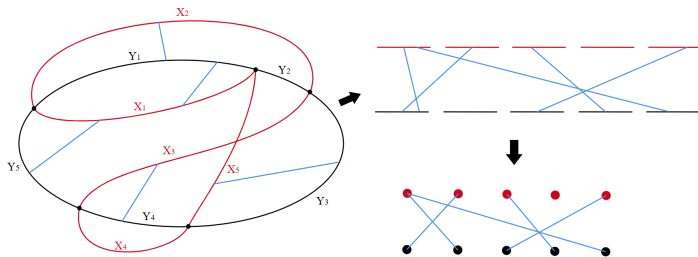
Menger: every  $(X, Y)$ -separator is large  
 $\iff$  many disjoint  $(X, Y)$ -paths.

# Proof ideas: an auxiliary graph

## Auxiliary graph $F = F(X, Y, \mathcal{P})$

Let  $\mathcal{P}$  be a collection of disjoint  $(X - Y, Y - X)$ -paths in  $G$ . The auxiliary bipartite graph  $F = F(X, Y, \mathcal{P})$  is constructed by:

- (1)  $V(F) = A \cup B$ ,  $A = \{x_1, \dots, x_m\}$  and  $B = \{y_1, \dots, y_m\}$ , where the vertex  $x_i$  (resp.  $y_i$ ) corresponds to the segment  $X_i$  (resp.  $Y_i$ ) in  $G$ .
- (2) For  $i, j \in [m]$ , add the edge  $x_i y_j$  for an  $(X_i, Y_j)$ -path in  $\mathcal{P}$ .

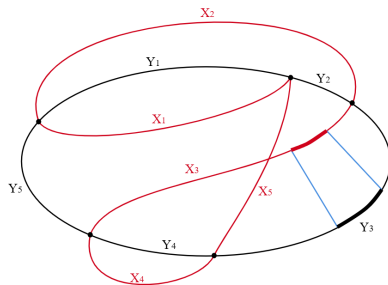


# Proof ideas

$F$  has  $2m$  vertices and  $|\mathcal{P}|$  edges, and " $H$ -free" for some  $H$ .

fact

$F$  contains no multi-edge.



# Proof ideas

$F$  is a simple graph with  $2m$  vertices and  $|\mathcal{P}|$  edges.

## Theorem

- (Chen et al., 1998)  $F$  is  $K_{3,257}$ -free.
- (Groenland et al., 2024)  $F$  is  $Q_3^+$ -free.
- (Ma, Z., 2025+) There are at most  $O(n)$  copies of  $K_{2,7}$  in  $F$ .

This implies  $|\mathcal{P}| = e(F) = O(\text{ex}(2m, K_{2,7})) = O(m^{3/2})$ .

# Proof ideas

A  $K_{2,7}$  in  $F$  means many  $K_{2,2}$  in  $F$ :

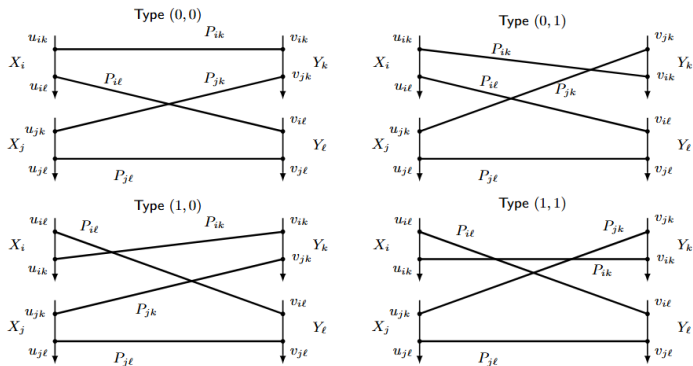


Figure: Classification for 4-cycles in  $F$ .

# Proof ideas for the main lemma

## Lemma

The graph  $F$  contains no 4-cycle of type  $(0, 0)$ .

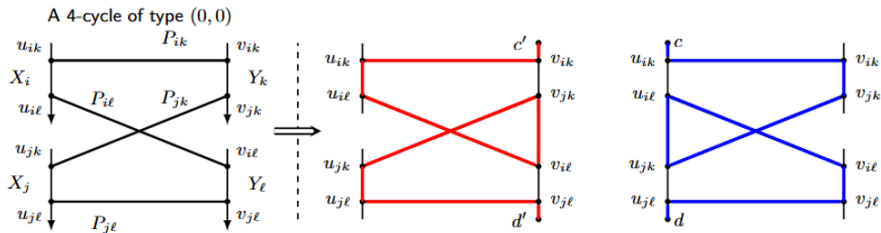


Figure: A 4-cycle of type  $(0, 0)$ .

# Proof ideas for the main lemma

## Lemma

*The graph  $F$  contains no two "interlacing" 4-cycles of type  $(1,0)$ .*

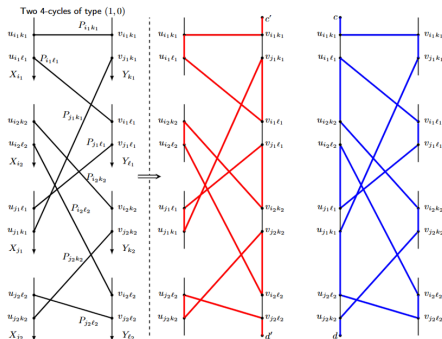


Figure: Two 4-cycles of type  $(1,0)$  in a special order.

# Proof ideas

## Main lemma 1

Let  $X$  and  $Y$  be two longest cycles in a 2-connected graph  $G$  with  $|V(X) \cap V(Y)| = m$ . Then there are at most  $O(m^{\frac{3}{2}})$  disjoint  $(X, Y)$ -paths in  $G$ .

Sketch: Suppose not, the auxiliary graph  $F$  contains many edges.

$\implies F$  contains many  $K_{2,7}$ 's.

$\implies$  Either a  $K_{2,2}$  of type  $(0, 0)$ , or two interlacing  $K_{2,2}$  of type  $(1, 0)$ , a contradiction.

# An intersection conjecture in vertex-transitive graphs

In cubic vertex-transitive graphs, Babai conjectured the following.

## Conjecture (Babai)

Any two longest cycles in a 3-connected cubic  $n$ -vertex graph must intersect in at least  $f(n)$  vertices, where  $f(n) \rightarrow \infty$  as  $n \rightarrow \infty$ .

We show that Babai's conjecture holds for the vertex-transitive case.

## Theorem (Ma, Z., 2025+)

*For each integer  $d \geq 2$ , every two longest cycles in a connected  $d$ -regular vertex-transitive graph on  $n$  vertices intersect in  $\Omega(\log_d^{1/3} n)$  vertices.*

## Main lemma 2 (Ma, Z., 2025+)

Every two longest cycles in a connected vertex-transitive graph on  $n$  vertices intersect in  $\Omega((\ln n / \ln \ln n)^{1/3})$  vertices.

# Ideas for intersection in vertex transitive graphs

## Theorem (Ma, Z., 2025+)

*For each integer  $d \geq 2$ , every two longest cycles in a connected  $d$ -regular vertex-transitive graph on  $n$  vertices intersect in  $\Omega(\log_d^{1/3} n)$  vertices.*

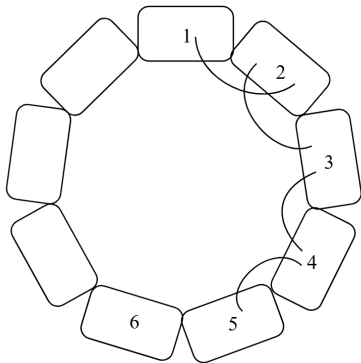
## Theorem (DeVos, Mohar, 2011)

*For every integer  $k \geq 1$ , if  $A$  is a separator of size  $k$  in a connected vertex-transitive graph  $G$  with  $\text{diam}(G) = \Omega(k^2)$ , then one of the following holds.*

- (1) Any two vertex sets on  $\Omega(k^3)$  are not separate by  $A$  ( $G$  is an expander).*
- (2)  $G$  "looks like" a cycle.*

# Ideas for intersection in vertex transitive graphs

$G$  "looks like" a cycle, where the two longest cycles  $X$  and  $Y$  separated by  $A$  are almost identical to the union of an interval of blocks.



**Figure:** Partition  $V(G)$  into several "stable" blocks under automorphisms

# Ideas for intersection in vertex transitive graphs

## Theorem (Ma, Z., 2025+)

*For each integer  $d \geq 2$ , every two longest cycles in a connected  $d$ -regular vertex-transitive graph on  $n$  vertices intersect in  $\Omega(\log_d^{1/3} n)$  vertices.*

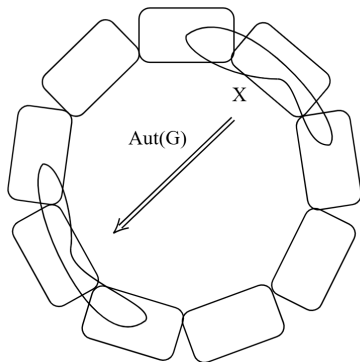
Sketch: Let  $X, Y$  be two longest cycles intersecting at  $m$  vertices.

**Expander-like case:** There is a separator on  $O(m^{3/2})$  vertices that separates  $X$  and  $Y$  (on  $\Omega(n^{1/2})$  vertices).

**Cycle-like case:** Let  $A$  be an  $(X, Y)$ -separator, then  $X \approx$  an interval of blocks.

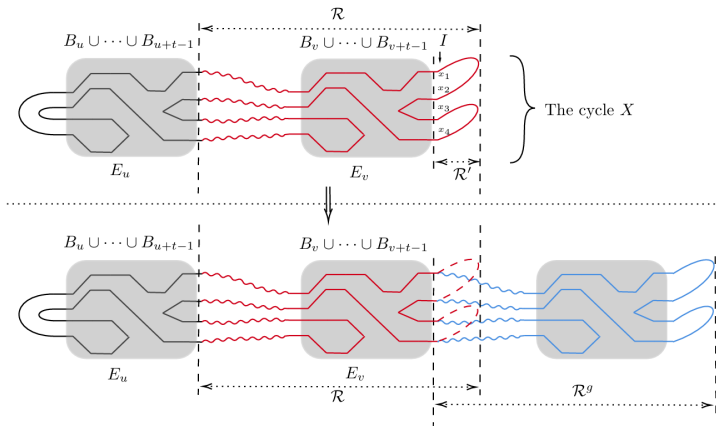
# Proof ideas

If  $X$  occupies less than half of the blocks, then an automorphism  $\sigma$  maps  $X$  to a longest cycle  $\sigma(X)$  disjoint from  $X$ .



# Proof ideas

Otherwise,  $X$  occupies sufficiently many blocks. There are two subintervals with the same local structure by the Pigeonhole Principle.



# Proof ideas

## Theorem (Ma, Z., 2025+)

*For each integer  $d \geq 2$ , every two longest cycles in a connected  $d$ -regular vertex-transitive graph on  $n$  vertices intersect in  $\Omega(\log_d^{1/3} n)$  vertices.*

## Main lemma 2 (Ma, Z., 2025+)

Every two longest cycles in a connected vertex-transitive graph on  $n$  vertices intersect in  $\Omega((\ln n / \ln \ln n)^{1/3})$  vertices.

# Thanks!